

Problem Chosen

C

2026
MCM/ICM
Summary Sheet

Team Control Number

2632449

EW

Summary

Keywords:

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1 Introduction

1.1 Problem Background

1.2 Restatement of the Problem

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2

2.1

1. 11
2. 22
3. 33

2.2 Notations

Some important mathematical notations used in this paper are listed in Table 1.

Table 1: Notations used in this paper

Symbol	Description
E	entropy

*There are some variables that are not listed here and will be discussed in detail in each section.

3 Estimating Latent Fan Votes: A Bayesian Inference Approach

The primary challenge in analyzing "Dancing with the Stars" (DWTS) lies in the fact that fan votes are latent variables, never publicly disclosed by the producers. However, the weekly elimination results and judges' scores provide a set of observable constraints. We propose a Bayesian framework to reverse-engineer the distribution of fan vote proportions.

3.1 Model Formulation and Notation

Let k_t be the number of contestants in week t . We define the following variables:

- $\mathbf{S}_t = \{s_{1,t}, s_{2,t}, \dots, s_{k_t,t}\}$: The set of normalized judges' scores for each contestant.
- $\boldsymbol{\theta}_t = \{\theta_{1,t}, \theta_{2,t}, \dots, \theta_{k_t,t}\}$: The latent proportion of fan votes, where $\sum_{i=1}^{k_t} \theta_{i,t} = 1$ and $\theta_{i,t} \geq 0$.
- y_t : The observed elimination result (the index of the contestant who was eliminated).

3.2 The Dirichlet Prior

The fan vote proportions reside on a $(k_t - 1)$ -dimensional simplex. The Dirichlet distribution is the natural conjugate prior for such compositional data. We assume:

$$\boldsymbol{\theta}_t \sim \text{Dirichlet}(\alpha_{1,t}, \alpha_{2,t}, \dots, \alpha_{k_t,t}) \quad (1)$$

where $\alpha_{i,t}$ represents the "prior popularity" of contestant i . In the absence of specific demographic data, we initially set $\alpha_{i,t} = \alpha_0 = 1$ (a flat prior). For iterative estimation across weeks, we update $\alpha_{i,t}$ based on the posterior mean of week $t - 1$, adjusted for recent performance.

3.3 Likelihood and Voting Constraints

The likelihood function $L(y_t | \boldsymbol{\theta}_t, \mathbf{S}_t)$ is defined by the specific voting rules of the season. Unlike traditional statistical models, the "observation" here is a ranking outcome, which acts as a hard constraint on the parameter space.

3.3.1 Percentage-Based System (Seasons 3–27)

In this system, the total score $V_{i,t}$ is the sum of percentages. Let $P_{i,t}^J = s_{i,t} / \sum s_{j,t}$ be the judges' score percentage. The elimination condition for contestant j is:

$$L(y_t = j | \boldsymbol{\theta}_t, \mathbf{S}_t) = \begin{cases} 1, & \text{if } P_{j,t}^J + \theta_{j,t} < P_{i,t}^J + \theta_{i,t} \quad \forall i \neq j \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

3.3.2 Rank-Based System (Seasons 1–2, 28–34)

In the rank-based system, let $R_{i,t}^J$ be the rank of contestant i based on judges' scores and $R_{i,t}^F$ be the rank based on fan votes $\theta_{i,t}$. The contestant with the highest sum of ranks (lowest performance) is eliminated:

$$L(y_t = j | \boldsymbol{\theta}_t, \mathbf{S}_t) = \begin{cases} 1, & \text{if } R_{j,t}^J + R_{j,t}^F > R_{i,t}^J + R_{i,t}^F \quad \forall i \neq j \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

3.4 Posterior Inference via MCMC

The posterior distribution $P(\boldsymbol{\theta}_t | y_t, \mathbf{S}_t) \propto L(y_t | \boldsymbol{\theta}_t, \mathbf{S}_t) \cdot \pi(\boldsymbol{\theta}_t)$ is analytically intractable due to the discontinuous nature of the likelihood. We employ Markov Chain Monte

Carlo (MCMC) sampling, specifically the Metropolis-Hastings algorithm, to sample from the constrained simplex space.

By generating $N = 10,000$ samples for each week, we estimate the mean fan vote $\hat{\theta}_{i,t} = E[\theta_{i,t}|\text{data}]$ and the associated uncertainty.

3.5 Measures of Consistency and Certainty

To evaluate the model, we define two metrics:

1. **Consistency Index (C):** The proportion of weeks where the maximum a posteriori (MAP) estimate of θ_t correctly identifies the eliminated contestant y_t .
2. **Certainty Score (σ):** Defined as the inverse of the Shannon Entropy of the posterior distribution:

$$\text{Certainty} = \left(- \sum_{i=1}^{k_t} \hat{\theta}_{i,t} \ln \hat{\theta}_{i,t} \right)^{-1} \quad (4)$$

High certainty indicates that the judges' scores were so disparate that only a very specific range of fan votes could have caused the observed outcome.

3.6

3.6.1

3.7 Differential Equation Model Based on System Dynamics

The overall model can be expressed as:

3.8

3.8.1 Shannon Diversity Index

4

4.1

4.2

- 1.
- 2.
- 3.

Table 2: Specified or Initial Values

Parameters or Variables	Values	Parameters or Variables	Values
r_P	0.1	β_A	0.1
k_G	1000	μ_A	0.05
α	0.05	β_B	0.1
r_W	0.5	μ_B	0.05
γ	0.05	P_0	100
η_I	0.0001	W_0	50
μ	0.05	I_0	30
μ'	0.02	A_0	1
η	0.05	B_0	1
μ_r	0.02	P_{r_0}	20

Table 3: Specified or Initial Values

Parameters or Variables	Values
η_I	0.005
η_{I_l}	0.005

*The values of the variables not listed are the same as those in Table 2.

5

5.1 Model Extension

5.2

5.2.1

5.2.2

5.2.3

- **Benefits:**
 - 1 **Ecological Benefits:**
 - 2 **Economic Benefits:**
 - 3 **Sustainability Benefits:**
- **Drawbacks**
 - 1 **Management Complexity:**
 - 2 **Ecological Risks:**

5.3

6 Strengths and Weaknesses

6.1 Strengths

- Logic:
- Creativity:
- Credibility:
- Stability:

6.2 Weaknesses

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References

- [1] information on <https://en.tutiempo.net/climate/cameroon.html>.

Letter

Report on Use of AI

1. OpenAI *ChatGPT* (<https://chat.openai.com/>)

Query1: Output:

Query2: Output:

Query3: Output: